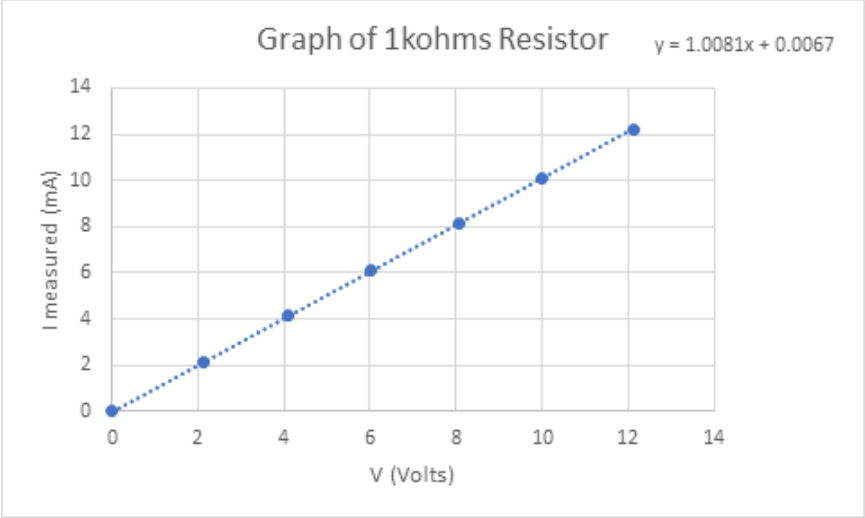
ECE 1101 Lab: Ohm's Law

Introduction: In this lab we are going to be using and verifying Ohm's Law. Ohm's Law states that we can get voltage by multiplying current by resistance. We will be using three different types of resistors 1k, 3.9k, and 8.2k. We will be measuring the resistance of all three types and then using that to get the power. We used I=V/R which was used to find current. Then we used P=V*I to get the power. Lastly, we used Dev=(It-Im/It)*100 to get the deviation.

Table 1: 1k Resistor

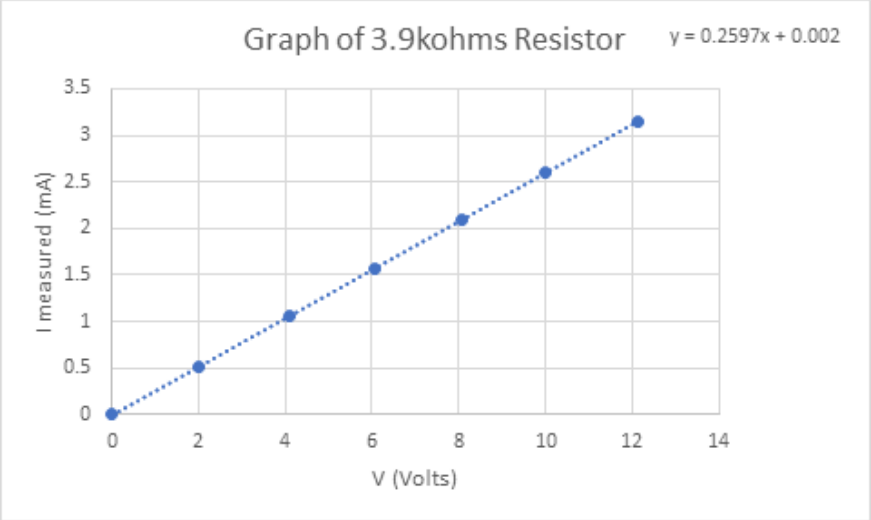
Volts(V)	I Theory	I measured	Deviation	Power
0	0mA	0.0007mA	undefined	0W
2.11	2.11mA	2.1351mA	-1.19%	4.50506W
4.09	4.09mA	4.1351mA	-1.11%	16.91460W
6.02	6.02mA	6.0784mA	-0.97%	36.59257W
8.08	8.08mA	8.1527mA	-0.90%	65.87381W
10.01	10.01mA	10.0948mA	-0.85%	101.04895 W
12.11	12.11mA	12.2128mA	-0.85%	147.90427 W



Explanation: Since the resistance is 1k and you find current by dividing voltage by resistance, we can assume that the current will be the same as the voltage. This was not the case though all the measured current values were in the similar range as the current theory but were not exact. This can be due to manufacturing error or rounding within production. Once we had out measured currents, we multiplied them by their specified voltage and got power. For our graph we got a slope of 1.0081 and a resistance of 0.992k Ohms.

Table 2: 3.9K Resistor

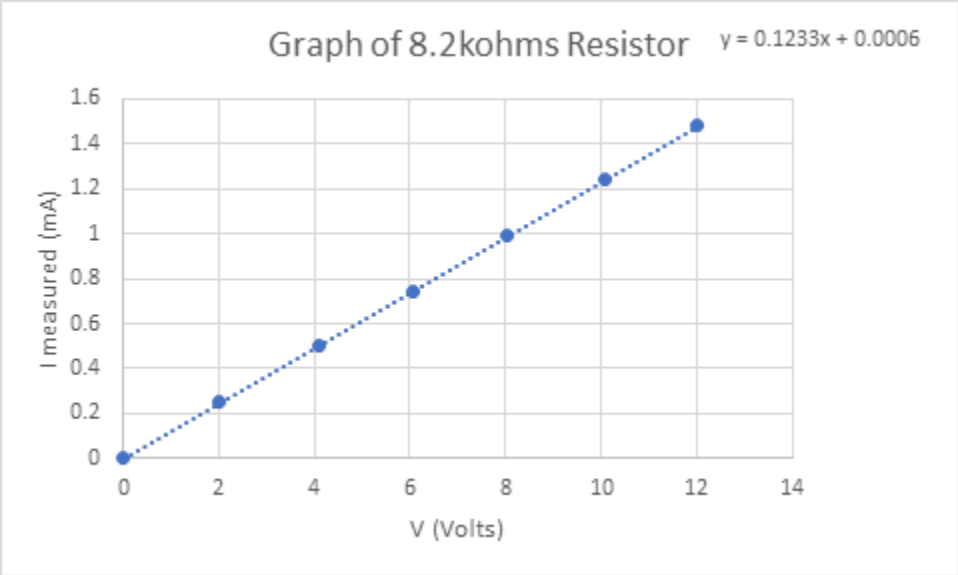
Volts(V)	I Theory	I measured	Deviation	Power
0	0mA	0.000179mA	undefined	0W
2	0.51282mA	0.52128mA	-1.65%	1.04258W
4.09	1.04615mA	1.06432mA	-1.74%	4.34242W
6.05	1.55128mA	1.57387mA	-1.46%	9.52191W
8.08	2.07179mA	2.0997mA	-1.35%	16.96557W
10	2.56410mA	2.5989mA	-1.36%	25.99W
12.14	3.11282mA	3.1541mA	-1.32%	38.28956W



Explanation: Unlike the last resistor, the resistance of this one is 3.9k which means we must divide our voltage values by this resistance. Our measured current values were extremely close to the theory current values but not quite spot on. Again, this can be due to manufacturing error or the resistors not being the exact spec they say they are. With our measured current values, we multiplied them by their voltage values and got power. For our graph, we got a slope of 0.2597 and a resistance of 3.820k Ohms

Table 3: 8.2k Resistor

Volts(V)	I Theory	I measured	Deviation	Power
0	0mA	0.000086mA	undefined	0W
2.01	0.24521mA	0.24844mA	-1.35%	0.49936 W
4.09	0.49878mA	0.50537mA	-1.32%	2.06696 W
6.05	0.73780mA	0.74650mA	-1.18%	4.51632 W
8.02	0.97804mA	0.98894mA	-1.11%	7.93129 W
10.06	1.22682mA	1.24066mA	-1.13%	12.48104W
12.01	1.46463mA	1.48090mA	-1.11%	17.78560W

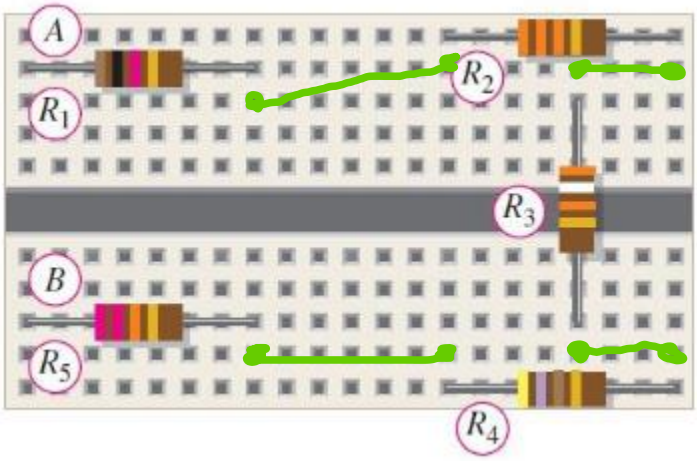


The resistance value of this resistor is 8.2k Ohms. We divided our voltages by 8.2k and got our current theory values. These measured current values were the closest to theory current values we got. With our measured current values, we multiplied them by the voltage to get our power values. Now although we were super close to our theoretical current values, we weren't exact which means there was some human error involved in the measuring process or the manufacturing process of the resistors. The slope of our graph is 0.1233 and has a resistance of 8.110k Ohms.

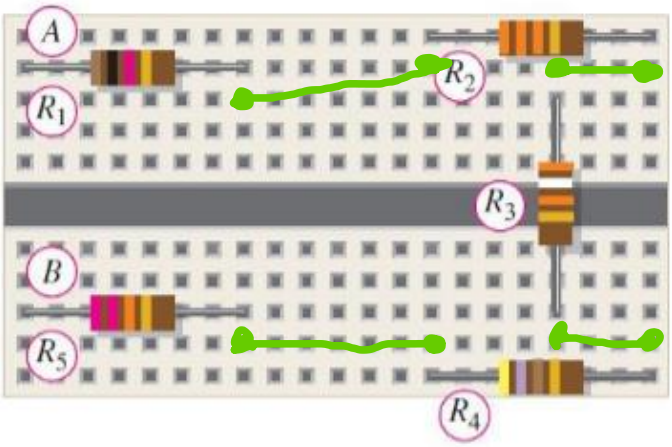
Conclusion: In this lab I learned how to physically measure the resistance value of a resistor. Then I refined my knowledge from lecture by practicing Ohm's Law (I=V/R) and using the power formula (P=V*I). The measured values are different from the theoretical values, most likely due to manufacturing error with the resistors. They could be intended to be made to have a resistance of x value but as long as they are close enough, a manufacturer will allow it to be sold still.

Post Lab:

1- Five resistors are positioned on a protoboard as shown in Figure 2. Wire them together in series so that, starting from the positive (+) terminal, R1 is first, R2 is second, R3 is third, and so on. Draw a schematic showing this connection.



2- Connect the resistors on the protoboard in Figure 3 in series, and determine the total resistance, RT, from the color codes



R1=1k Ohms, R2=33k Ohms, R3=39k Ohms, R4=470 Ohms or 0.47k Ohms, and R5=22k Ohms. Total Resistance = R1+R2+R3+R4+R5 = 1k+33k+39k+0.47k+22k=95.47k Ohms.